METRIC
DOD-C-24676
7 December 1987

MILITARY SPECIFICATION

COPPER-NICKEL ALLOY BAR, ROD AND FORGED SHAPES, HIGH STRENGTH (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of the Defense.

1. SCOPE

1.1 Scope. This specification covers copper-nickel alloys with 58,000 minimum yield strength in wrought bars, rod and forged shapes.

1.2 Classification. Copper-nickel alloy covered by this specification shall be of the forms specified (see 6.2.1).

Forms:
(1) Reforging stock
(2) Bar
(3) Rod
(4) Forged shapes

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY
MIL-C-3993 - Copper and Copper-Base Alloy Mill Products; Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N4182 AREA FORG

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STANDARDS

FEDERAL

FED-STD-146 - Tolerances for Copper and Copper Base Alloy Mill Products.
FED-STD-185 - Identification Marking of Copper and Copper Base Alloy Mill Products.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-271 - Nondestructive Testing Requirements for Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)
E 8 - Standard Methods of Tension Testing of Metallic Materials. (DoD adopted)
E 75 - Standard Methods for Chemical Analysis of Copper-Nickel and Copper-Nickel-Zinc Alloys.
E 340 - Standard Method for Macroetching Metals and Alloys. (DoD adopted)
E 478 - Standard Methods for Chemical Analysis of Copper Alloys.

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and manufacture. The material shall be of such quality and purity that the finished product shall have the properties and characteristics specified herein. The material shall be manufactured by hot working, cold working, or both, and finished by such cold working, annealing, or heat treatment and straightening as may be necessary to meet the properties specified.

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3.1.1 When specified (see 6.2.1), forgings shall be shaped in such a manner that flow lines will lie in the direction of the major applied stress and not cut across any pressure boundary. Flow line directions shall be as shown on the forging sketch.

3.2 Recovered materials. Unless otherwise specified herein, all material covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials.

3.3 Chemical composition. The chemical composition of the alloy shall conform to table I.

TABLE I. Percent chemical composition.

Element	Minimum (percent)	Maximum (percent)
Aluminum	1.0	2.0
Iron	0.7	1.2
Nickel and cobalt	13.5	16.5
Manganese	3.5	5.5
Chromium	---	0.5
Copper and silver	Remainder[1]	
Tin	Trace elements[2]	0.10
Lead	---	0.02
Silicon	---	0.15
Magnesium	---	0.05
Sulphur	---	0.15
Phosphorus	---	0.01
Zinc	---	0.20
Carbon	---	0.05
Total trace elements	---	0.30

[1] Copper may be taken as the difference between the sum of all the elements analyzed and 100 percent. When all the elements in table I are analyzed their sum shall be 99.7 percent minimum.

[2] Each trace element listed shall not exceed value shown. The sum of all trace elements listed shall not exceed 0.30 percent.

3.4 Mechanical properties. Mechanical properties of the material shall conform to table II.

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TABLE II. Mechanical properties.

Property	Requirement
Tensile strength, lb/in^2 (MPa) (minimum)	103,000 (710)
Yield strength at 0.2 percent offset, lb/in^2 (MPa) (minimum)	58,000 (400)
Elongation (in 2 inches or 4 times diameter) percent (minimum)	18 (18)

3.5 Forging sketch. When specified in the contract or order, a sketch shall be prepared for forgings other than solid or hollow rods or bars, rings, and disks or pancakes (see 6.2.2).

3.6 Dimensional. The dimensions shall be as specified (see 6.2.1), in the form of a sketch for forgings or numerical dimensions for standard cross-sections. Sketches and ordering data for nonsymmetrical forgings shall include tolerances.

3.6.1 Tolerances. Dimensional tolerances for rod and bars shall be in accordance with FED-STD-146.

3.7 Weld repair. Brazing or weld repair shall not be permitted.

3.8 Soundness. The material shall be free of cracks, laminations, seams, tears, cold shuts, laps, and scale to the maximum extent possible as determined by visual inspection or by the specified nondestructive tests. Surface imperfections such as handling marks, light mandrel and die or roll marks will not be considered injurious defects, provided the imperfections do not detract from the product's end use.

3.8.1 Nondestructive inspection. The nondestructive test method shall be designated and acceptance criteria shall be as specified (see 6.2.1).

3.8.2 Defect removal. The removal of surface defects shall be accomplished by grinding, machining, or filing with a clean file provided such removal does not prevent the item from meeting the requirements specified herein. Repaired areas shall be faired smoothly into the surrounding material so that the bottom radius of the repaired area equals at least three times the depth of repair. The removal of the defect shall be verified by the same method used to detect the defect.

3.9 Material identification. Identification marking shall be in accordance with FED-STD-185 and shall include the following information:

(a) Manufacturer's name, trademark, or symbol.
(b) Heat or lot number, or a unique number identifying the heat or lot.
(c) Alloy identification symbol.
(d) Specification number and revision letter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. When specified in the contract or order, a certificate of compliance shall be prepared for each lot of material offered for delivery (see 6.2.2).

4.2 Lot.

4.2.1 Bar, rod and forgings weighing 250 pounds (115 kilograms (kg)) or less. A lot shall consist of material having a total weight of 10,000 pounds (4500 kg) or less, of the same design, and from the same heat treatment batch. A lot shall be from the same heat and shall be identified with respect to heat.

4.2.2 Bar, rod and forgings weighing over 250 pounds (115 kg). A lot shall consist of each individual forging. Each forging shall be identified with respect to heat.

4.3 Sampling.

4.3.1 Samples shall be taken from integral prolongations, from an actual forging from the lot, or from the separately forged test specimens experiencing the same amount of work and reduction.

4.3.2 Sampling for chemical analysis.

4.3.2.1 Bar and rod. One sample shall be taken from each lot.

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4.3.2.2 Forgings.

4.3.2.2.1 Forgings less than 250 pounds (115 kg) in as-forged weight. Samples shall be taken from two different production forgings selected at random from each lot.

4.3.2.2.2 Forgings weighing 250 pounds (115 kg) or more in as-forged weight. Each forging shall be sampled.

4.3.3 Sampling for mechanical tests and macroexamination.

4.3.3.1 Bar and rod. One sample shall be selected for each test from the lot.

4.3.3.2 Forgings.

4.3.3.2.1 Forgings 250 pounds (115 kg) and less. Two samples shall be taken from each lot to provide sufficient material for two sets of mechanical tests. Prolongations shall be provided on at least 20 percent of the forgings unless trepanning is used, or the manufacturer may provide extra forgings for testing.

4.3.3.2.2 Forgings greater than 250 pounds (115 kg). Sufficient samples shall be taken from each forging to provide two sets of mechanical test data. Prolongations shall be provided on each forging unless trepanning is used.

4.3.4 Forging multiples. Where forgings are made in multiples from a single forging; that is, forged in one piece and machined apart, individual tests of each forging need not be made, but tests of the large forging shall govern.

4.4 Tests.

4.4.1 Chemical analysis. From each sample (see 4.3.2), not less than 2 ounces (57 grams) of clean millings, drillings or chippings shall be obtained for chemical analysis. Either the wet chemical method of ASTM E 75 or ASTM E 478, or the spectrographic method may be used. In the case of dispute, the wet chemical method shall be the basis for acceptance. When analyzing for phosphorus and sulphur content, each heat or ingot shall be sampled. If any sample fails to meet the requirements of table I, this shall be cause for rejection of the lot represented by the sample.

4.4.1.1 Acceptance criteria. If any sample fails to conform to the chemical composition of table I, the heat represented by the sample shall be rejected.

4.4.2 Tension tests. Each of the specimens selected in accordance with 4.3.3 shall be subjected to the tension test as follows:

(a) The tension tests shall be in accordance with ASTM E 8. The largest round tensile test specimens shall be used as required by ASTM E 8.

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(b) The longitudinal axis of tension test specimens shall be parallel to the direction of forging.

4.4.2.3 Acceptance criteria. If any test fails to meet the requirements of table II, the lot represented by that specimen shall be rejected. Specimen replacement in accordance with ASTM E 8 is permitted.

4.4.3 Macroetch test. When a specific grain flow is required (see 3.1.1 and 6.2.1), forgings shall be prepared and etched in accordance with ASTM E 340 and examined for conformance to 3.1.1.

4.4.3.1 Acceptance criteria. If the grain flow pattern fails to meet the requirement of 3.1.1, the lot represented by the sample shall be rejected.

4.4.4 Ultrasonic inspection.

4.4.4.1 General. Each bar, rod or forging shall be ultrasonically inspected for soundness. Testing shall be performed by the longitudinal or shear wave technique as specified herein and in accordance with MIL-STD-271.

4.4.4.2 Calibration.

4.4.4.2.1 Shear wave. The shear wave test shall be calibrated on a notch cut into the inside and outside surfaces. Depth of the notches shall be 5 percent of the material thickness or 1/4 inch (6.35 millimeters (mm)), whichever is less, and their length shall not exceed 1 inch. Notches shall be made within the piece to be examined or in a separate defect-free specimen of the same size, shape, material, and condition. The notch producing the smaller amplitude response shall be used as the evaluation reference.

4.4.4.2.2 Longitudinal wave. The longitudinal wave test shall be calibrated on a flat-bottomed reference hole of a given diameter in accordance with table III for specified material thickness drilled either into the piece to be tested or into a separate defect-free specimen of the same size, shape, material, and condition. Holes shall be drilled to mid-section in material up to 1-1/2 inches (38 mm) thick and at least 3/4 inch (19 mm) deep, but no greater than mid-section in material over 1-1/2 inches thick, and shall be normal to the surface. The ultrasonic test instrument shall be adjusted so that the response from the reference hole shall be not less than 25 percent and not more than 75 percent of screen height.

TABLE III. Bars, rods and forgings.

Material thickness, (inches) (mm)	Hole diameter, (inches) (mm)
Up to and including 6 (152)	1/8 (3.2)
Over 6 and including 16 (406)	1/4 (6.4)

4.4.4.2.3 Recalibration. During quality conformance inspection, any realignment of the search unit with respect to the material, or any change in search unit, couplant, instrument settings or scanning speed from that used for calibration, shall require recalibration. Recalibration shall be performed

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at least once per 8-hour shift. If a recheck of the calibration standard indicates that recalibration is required, all items tested since the last instrument check shall be retested.

4.4.4.2.3.1 Procedure. Simple forged shapes shall be tested in accordance with the following procedures. Shapes other than those listed below shall be tested to the extent specified in the approved forging sketch (see 3.5).

(a) Solid cylindrical forgings shall be tested using the longitudinal wave technique. The scanning path shall be circumferential or helical with the beam directed along a radius of the cylinder.
(b) Solid non-cylindrical forgings shall be tested using the longitudinal wave technique through each pair of parallel sides.
(c) Ring and hollow round forgings shall be tested using the shear wave technique. The angle used shall ensure reflection from the internal surface. The search unit shall be fitted with a wedge or shoe machined to fit the curvature of the piece being inspected. The forgings shall also be inspected with a longitudinal wave test from the external circumferential and end surfaces.
(d) Disk or pancake forgings shall be tested using the longitudinal wave technique from both parallel surfaces.

4.4.4.3 Acceptance criteria.

4.4.4.3.1 Shear wave. Any material which produces indications equal to or greater than the response from the reference notch shall be rejected.

4.4.4.3.2 Longitudinal wave. Any material which produces indications equal to or greater than the response from the reference hole, or which produces a complete loss of back reflection, shall be rejected.

4.4.4.3.3 Reference notch removal. If reference notch or flat-bottomed holes are made in the material to be tested, they shall be so located that their subsequent removal will not impair the suitability of the material for its intended use.

4.4.5 Liquid penetrant inspection.

4.4.5.1 Procedure. Each bar, rod, or forging shall be liquid penetrant inspected in accordance with MIL-STD-271.

4.4.5.2 Acceptance criteria. Liquid penetrant indications shall be considered to be defects and shall be evaluated in accordance with the following criteria: all surfaces shall be free of linear indications and of linearly disposed rounded indications when there are four or more such rounded indications in a line, and each is separated from the adjacent indications by less than 1/16 inch (1.60 mm). Rounded indications are any indications which are circular or elliptical, with the long axis less than twice as long as the other axis, and with no sharp corner. Rounded indications which are not linearly disposed shall be evaluated in accordance with table IV. Material exhibiting rounded indications in excess of those specified in table IV shall be rejected.

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TABLE IV. Allowable nonlinear rounded indications.

Maximum size (inches) (mm)	Cumulative number per square inch (per 100 cm^2)
Up to 1/32 (0.80), inclusive	20 (80)
Over 1/32 (0.80) to 1/16 (1.60), inclusive	10 (40)
Over 1/16 (1.60)	0 (0)

4.4.6 Visual and dimensional examination. Each piece of material shall be examined to determine conformance with the visual and dimensional requirements of this specification and the drawing or purchase order.

4.5 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation and packing.

5.1.1 Military. Unless otherwise specified (see 6.2.1), material shall be preserved level A or C and packed level A, B or C in accordance with MIL-C-3993.

5.1.2 Commercial. When specified (see 6.2.1), material shall be preserved and packed in accordance with ASTM D 3951.

5.2 Marking. In addition to any special marking required (see 6.2.1), interior and exterior containers packed to levels A, B, or C shall be marked in accordance with MIL-C-3993. Commercial packs shall be marked in accordance with ASTM D 3951. Bar coding shall be applied to all containers in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Copper-nickel alloys of this composition are intended for use in corrosive environments such as seawater and high shock application where high strength is required.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Form to be furnished (see 1.2).
(c) Whether a specific grain flow is required (see 3.1.1 and 4.4.3).

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(d) The finished dimensions and tolerances of each forging (see 3.6).
(e) The nondestructive test method and acceptance criteria to be used (see 3.8.1).
(f) Level of preservation and packing required (see 5.1.1).
(g) When commercial packaging is required (see 5.1.2).
(h) Special marking required (see 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.5	Forging manufacturing sketch	DI-FORG-80412	----
4.1.2	Certification data for level I material	UDI-T-23191	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Forging sketch. The sketch prepared in accordance with 3.5 should be submitted to and approved by the contracting activity prior to starting forging operations.

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6.4 Subject term (key word) listing.

Forging
Forging, ultrasonic inspection
Liquid penetrant inspection, forgings
Macroetch inspection, forgings

Custodians:
Army - MR
Navy - SH
Air Force - 20

Review activities:
Army - ME, GL, EA, AR

Preparing activity:
Navy - SH
(Project FORG-0137)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-C-24676	Copper-Nickel Alloy Bar, Rod and Forged Shapes, High Strength

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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